CODING COMFORT

AI-Assisted Community Mental Health Screening for Underserved Filipino Populations

Project Overview

Coding Comfort is a nursing thesis/research prototype designed to support emotional self-reflection, standardized mental health screening, and connection to appropriate support services among underserved Filipino communities.

The platform must have three major functional components:

1. **DASS-21 Screening:** The main and only standardized mental health screening instrument.
2. **AI-Assisted Reflection:** Explains the user's screening results using supportive, safe, and non-diagnostic language.
3. **Referral Guidance and Human Oversight:** Provides appropriate recommendations based on screening results and safety concerns while ensuring that AI does not independently diagnose or make clinical decisions.

Supplementary wellness features may include free-form emotional expression, AI-assisted reflection, guided self-care activities, and mental health support resources. These features must remain separate from the DASS-21 screening instrument.

Global Requirements

Navigation

Provide:

- Home
- DASS-21 Screening
- My Results
- Self-Care
- Support

ENGLISH AND FILIPINO LANGUAGE TOGGLE

Use natural, accessible Filipino rather than overly formal or technical Tagalog.

Use the exact supplied Filipino DASS-21 wording in Section 6. Do not automatically replace it with machine-generated translations.

Allow language switching without clearing questionnaire answers, journal drafts, or current progress.

Implement translations through a centralized translation dictionary or localization system.

Landing Page

Create a welcoming landing page.

Main title: Coding Comfort

Tagline: Your feelings matter. Take a moment for yourself.

Introduction:

"Some days feel lighter than others. Sometimes we carry worries about family, school, work, finances, relationships, and responsibilities without enough time to understand how we feel. Coding Comfort gives you a space to check in with yourself, express your feelings, and explore small steps toward emotional well-being."

Display a short explanation:

"Coding Comfort uses the DASS-21 to help you understand your reported experiences of depression, anxiety, and stress. It also provides supportive reflections and guidance toward appropriate mental health resources. It is not a diagnostic tool."

Main feature cards

Create 4 prominent feature cards.

Card 1: DASS-21 Mental Health Screening

Description:

"Take a standardized 21-item questionnaire to reflect on experiences related to depression, anxiety, and stress over the past week."

Button: Start DASS-21

This must be the most prominent feature on the homepage.

Card 2: Understand My Results

Description:

"Receive an easy-to-understand explanation of your DASS-21 results and explore appropriate next steps."

Button: Understand My Results

Only make this available after completing the screening.

Card 3: Self-Care Corner

Description:

"Explore breathing exercises, grounding activities, guided journaling, and small self-care steps."

Button: Explore Self-Care

Card 4: Mental Health Support

Description:

"Find reliable support resources, referral guidance, and information about seeking professional assistance."

Button: Find Support

All five cards must navigate to functional pages or sections.

CONSENT AND PRIVACY

Before users begin the DASS-21 screening or any activity that collects or processes research responses, present an informed-consent screen.

Heading: Before We Begin

Explain that:

- Participation is voluntary.
- Participants may stop at any time.
- Coding Comfort is a research prototype.
- DASS-21 is a screening instrument, not a diagnostic tool.
- AI may process information voluntarily submitted by the participant.
- Participants should avoid including names, addresses, phone numbers, or other unnecessary identifying information in free-text entries.
- The application is not an emergency service.

Provide English and Filipino versions.

Add a required checkbox:

"I have read and understood the information above and voluntarily agree to continue."

The Continue button must remain disabled until consent is provided.

If consent is declined, do not collect questionnaire answers or journal entries.

Clearly disclose which information is transmitted to Claude and which information is stored for research.

Obtain any additional consent required for AI processing, retention, or secondary research use.

Do not promise complete anonymity or confidentiality unless the actual system and approved research procedures support those claims.

Keep research participation consent separate from optional wellness features when necessary.

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6. MAIN FEATURE: DASS-21 MENTAL HEALTH SCREENING

This is the central and most important feature of Coding Comfort.

Use the **Depression, Anxiety and Stress Scale – 21 Items (DASS-21)** as the sole standardized mental health screening questionnaire.

A. DASS-21 introduction

Title: Depression, Anxiety and Stress Scale (DASS-21)

Subtitle: Understand your emotional experiences over the past week.

English description:

"The DASS-21 is a 21-item self-report questionnaire that measures three related emotional states: depression, anxiety, and stress. It asks about experiences during the past week and helps provide a clearer understanding of your emotional well-being."

Filipino description:

"Ang DASS-21 ay isang questionnaire na binubuo ng 21 katanungan na sumusukat sa tatlong magkakaugnay na emosyonal na kalagayan: depresyon, pagkabalisa, at stress. Tinatanong nito ang mga karanasang naramdaman mo sa nakalipas na isang linggo upang makatulong na mas maunawaan ang iyong emosyonal na kalagayan."

Explain that:

- The questionnaire contains 21 statements.
- It measures depression, anxiety, and stress separately.
- It refers to experiences during the past week.
- There are four response options.
- There are no right or wrong answers.
- Participants should answer based on their actual experiences.
- DASS-21 is a screening instrument, not a standalone diagnosis.
- About 3–5 minutes.

Button: Start DASS-21 Screening

B. DASS-21 instructions

English:

"Please read each statement and select the response that indicates how much the statement applied to you over the past week."

Filipino:

"Basahin ang bawat pahayag at piliin ang sagot na nagpapakita kung gaano ito naaangkop sa iyo sa nakalipas na isang linggo."

C. Filipino response options

Use the following supplied Filipino response options:

Score	Filipino response
0	Hindi nangyayari / nagaganap sa akin

Score	Filipino response
1	Paminsan-minsan nangyayari / nagaganap sa akin
2	Pangkaraniwang nangyayari / nagaganap sa akin
3	Madalas na nangyayari / nagaganap sa akin

Use these response options consistently for all 21 items.

The English version must use the corresponding standard DASS-21 response scale:

Score	English response
0	Did not apply to me at all
1	Applied to me to some degree, or some of the time
2	Applied to me to a considerable degree, or a good part of the time
3	Applied to me very much, or most of the time

Do not change the numerical scoring.

D. The 21 Filipino DASS-21 items

Use the following Filipino questionnaire wording supplied by the research team.

Important: Preserve the supplied wording during implementation. Do not silently rewrite, remove, add, or rearrange items. Flag any potential translation or wording issues for research-team verification before actual participant deployment.

No.	Filipino DASS-21 statement
1	Nahihirapan akong kumalma.
2	Batid ko ang panunuyo ng aking bibig.
3	Hindi ko man lang maranasan ang makaramdam ng mabuti.
4	Nakakaranas ako ng hirap na paghinga (hal. sobrang bilis ng paghinga, kawalan ng hininga kung hindi aapuhin ito).
5	Nahihirapan akong magkusa na gumawa.
6	Masyado akong nagiging intimidida sa mga sitwasyon.
7	Nakaranas ako ng panginginig (hal. sa kamay).
8	Naramdaman ko na masyado akong gumagamit ng aking nervous energy.
9	Nangangamba ako sa mga sitwasyong kung saan maaari akong matuliro at makagawa nang hindi matino.

No.	Filipino DASS-21 statement
10	Nararamdaman ko na wala naman akong inaasahan.
11	Madali akong mainis o hindi mapakali.
12	Nahihirapan akong magrelaks / mamahinga.
13	Nakaramdam ako ng lungkot at panghihina ng loob.
14	Hindi ko pinapansin ang bagay na nakakahadlang sa aking ginagawa.
15	Naramdaman kong muntik na akong matuliro.
16	Hindi ko nagagawang maging magilas / aktibo sa kahit anong bagay.
17	Naramdaman kong wala akong kwenta.
18	Naramdaman kong madali akong magdamdam o mairita.
19	Batid ko ang galaw ng aking puso kahit hindi ako kumikilos (hal. pagbilis ng pintig, pagkawala ng pintig ng puso).
20	Natatakot ako ng walang kadahilanan.
21	Naramdaman kong walang kahulugan ang buhay.

E. Questionnaire interface

Display one question at a time.

Each question screen must include:

- DASS-21 title.
- Question number.
- Question statement.
- Four large selectable response cards.
- Previous and Next buttons.
- Progress indicator.
- Clearly visible selected-answer state.

Require an answer before proceeding.

Preserve responses when users navigate backward or change languages.

Display progress in a simple format:

"Question 7 of 21"

Include a visual progress bar. “Sun Arc Progress feature”

F. DASS-21 scoring

Calculate three separate subscale scores.

Each subscale contains seven items.

Depression items:

3, 5, 10, 13, 16, 17, 21

Anxiety items:

2, 4, 7, 9, 15, 19, 20

Stress items:

1, 6, 8, 11, 12, 14, 18

Each response is scored from 0 to 3.

Calculate the raw score for each subscale separately.

Each raw subscale score ranges from 0 to 21.

For conventional DASS-21 interpretation, multiply each raw subscale score by 2.

Final interpreted scores range from 0 to 42 for each subscale.

Implement the following conventional severity categories using the doubled scores:

Severity	Depression	Anxiety	Stress
Normal	0–9	0–7	0–14
Mild	10–13	8–9	15–18
Moderate	14–20	10–14	19–25
Severe	21–27	15–19	26–33
Extremely Severe	28+	20+	34+

Important scoring requirements:

- Do not combine the three subscale scores into one overall score.
- Do not use one subscale's severity category to interpret another.
- Calculate scores deterministically in the application backend or a validated scoring module.
- Require all 21 items to be answered before calculating the final results.
- Ensure the displayed score corresponds to the correct scoring method.

These categories describe conventional DASS-21 symptom severity classifications. They are not clinical diagnoses.

G. DASS-21 results page

Title: Your DASS-21 Screening Results

Create three separate result cards:

1. Depression
2. Anxiety
3. Stress

Each card must display:

- Subscale name.
- Final score.
- Conventional severity category.
- Simple visual score indicator.
- Short, understandable explanation.

For example:

Depression

"Your responses fall within the moderate symptom range for the depression subscale. This indicates that you reported experiences associated with depressive symptoms during the past week. It does not mean that you have been diagnosed with depression."

Use corresponding explanations for Anxiety and Stress.

Display the following disclaimer:

"The DASS-21 is a screening instrument that measures reported symptoms of depression, anxiety, and stress. It does not establish a mental health diagnosis. Your results should be interpreted in context, and a qualified health professional can provide further assessment when needed."

Provide a natural Filipino translation.

Include the following buttons:

- Help Me Understand My Results
- Find Appropriate Support
- Explore Self-Care
- Return to Home

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7. AI COMPONENT: EXPLANATION AND PERSONALIZED REFLECTION

The AI component must explain screening results in a supportive, understandable, and non-diagnostic manner.

The AI must:

1. Explain the meaning of each subscale in simple language.
2. Explain the conventional severity category without presenting it as a diagnosis.
3. Identify general patterns supported by the participant's actual responses.
4. Explain that depression, anxiety, and stress are separate but potentially related emotional experiences.
5. Provide supportive and nonjudgmental language.
6. Suggest one or two realistic, low-risk self-care steps when appropriate.
7. Recommend professional assessment when indicated.
8. Provide appropriate referral guidance.
9. Respond in the user's selected language.
10. Avoid assumptions not supported by the questionnaire responses.

Example:

"Your responses suggest that you have experienced some difficulties related to stress during the past week. Managing responsibilities, daily pressures, and personal concerns can sometimes feel overwhelming. It may help to take short breaks, identify one manageable task, or talk with someone you trust. If these experiences continue or affect your daily activities, consider speaking with a qualified health professional."

Do not use identical generic responses for every participant.

Do not:

- Diagnose depression, anxiety disorders, or other conditions.
- Claim clinical certainty.
- Prescribe medication.
- Recommend stopping prescribed treatment.
- Make independent clinical decisions.
- Predict future mental health conditions.
- Declare someone mentally healthy based on a low score.
- Tell users that AI can replace professional assessment.

The AI must clearly state that its explanation is informational and supportive.

AI follow-up actions

After the explanation, provide optional buttons:

- Explain My Results Further.
- Suggest a Small Next Step.
- Recommend Where I Can Seek Help.
- Try a Calming Exercise.
- Finish Reflection.

Limit follow-up exchanges and provide a clear exit option.

Keep AI explanations separate from the original DASS-21 score calculations.

8. REFERRAL GUIDANCE AND HUMAN OVERSIGHT

This is a core component of Coding Comfort.

The application must help users understand where and how to seek appropriate assistance based on their screening results and expressed concerns.

Referral guidance must not be treated as an automated clinical referral or diagnosis.

A. Referral recommendations based on DASS-21 results

Use the three separate subscale interpretations to provide general guidance.

For example:

- Normal: Encourage continued self-care and awareness of emotional well-being.
- Mild: Offer practical self-care suggestions and information about available support.
- Moderate: Encourage consideration of consultation with a qualified mental health professional, especially when experiences persist or affect daily functioning.
- Severe: Strongly encourage timely professional assessment and connection with appropriate support services.
- Extremely Severe: Prioritize timely professional evaluation and provide accessible support information.

These are general support recommendations, not mandatory clinical triage categories.

Do not assume that a person with a normal score does not need assistance.

Do not automatically classify someone as being in immediate danger based on a DASS-21 score.

B. Human oversight

Clearly explain:

"AI-generated reflections are provided to support understanding and self-reflection. They do not replace professional judgment. Decisions regarding diagnosis, treatment, and clinical care must be made by qualified healthcare professionals."

The AI must never independently:

- Diagnose a mental health disorder.
- Determine clinical risk.
- Prescribe medication.
- Make treatment decisions.
- Claim that a referral has been completed.
- Contact a healthcare provider without an actual authorized integration and appropriate consent.

10. FEATURE: SELF-CARE CORNER

Create an interactive self-care area with the following sections.

A. Breathing Exercise

Create a guided breathing exercise using a slowly expanding and contracting circle.

Include:

- Start.
- Pause.
- Resume.
- Restart.
- Exit.
- Adjustable duration.

Instruction:

"Breathe in slowly, then breathe out comfortably. You do not need to force your breathing."

Provide Filipino instructions.

Respect reduced-motion preferences.

Do not imply that breathing exercises resolve every emotional concern.

B. Grounding Exercise

Provide a guided 5-4-3-2-1 grounding exercise:

- Notice five things you can see.
- Notice four things you can feel.
- Notice three things you can hear.
- Notice two things you can smell.
- Notice one thing you can taste, if comfortable.

Allow users to proceed at their own pace.

Provide Filipino translations.

C. Small-Step Planner

Allow users to select one manageable action:

- Take a short break.
- Drink a glass of water.
- Talk with someone I trust.
- Spend a few minutes outside.
- Organize one small task.
- Ask someone for help.
- Make time for rest.

Allow a custom action.

Provide a simple completion state without guilt-inducing streaks, competition, or shaming notifications.

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11. FEATURE: MENTAL HEALTH SUPPORT CENTER

Create a dedicated support page containing:

- Mental health support organizations.
- Guidance on seeking professional help.
- Suggestions for starting a conversation with a trusted person.
- Information about when professional support may be useful.
- Crisis and emergency instructions.

Create resource configuration entries for:

- National Center for Mental Health (NCMH).
- Hopeline Philippines.
- In Touch Community Services.

Do not invent hotline numbers, operating hours, or service availability.

Verify current contact information through official sources before deployment.

Each resource card should include, when verified:

- Organization name.
- Official phone number.
- Available hours.
- Call or text options.
- Official website.
- Click-to-call functionality on mobile where supported.

Provide a prominent button:

I Need Help Now

If the user indicates immediate danger, advise contacting local emergency services or going to the nearest emergency department.

Do not promise emergency monitoring or automatic human intervention unless the application genuinely provides it.

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15. CONTACT THE CODING COMFORT TEAM

Include a dedicated "Contact the Team" section that allows users to reach the Coding Comfort research and development team for questions, technical concerns, feedback, or inquiries about the platform.

Contact Information

Team Email: codingcomfort.team@gmail.com

Section Title: Get in Touch With Us

Description: Have questions, concerns, or suggestions about Coding Comfort? Our team is here to help. Feel free to contact us through our official email address.

Functional Requirements

Display the official email address prominently: codingcomfort.team@gmail.com.

Include a clickable "Send Us an Email" button that opens the user's default email application